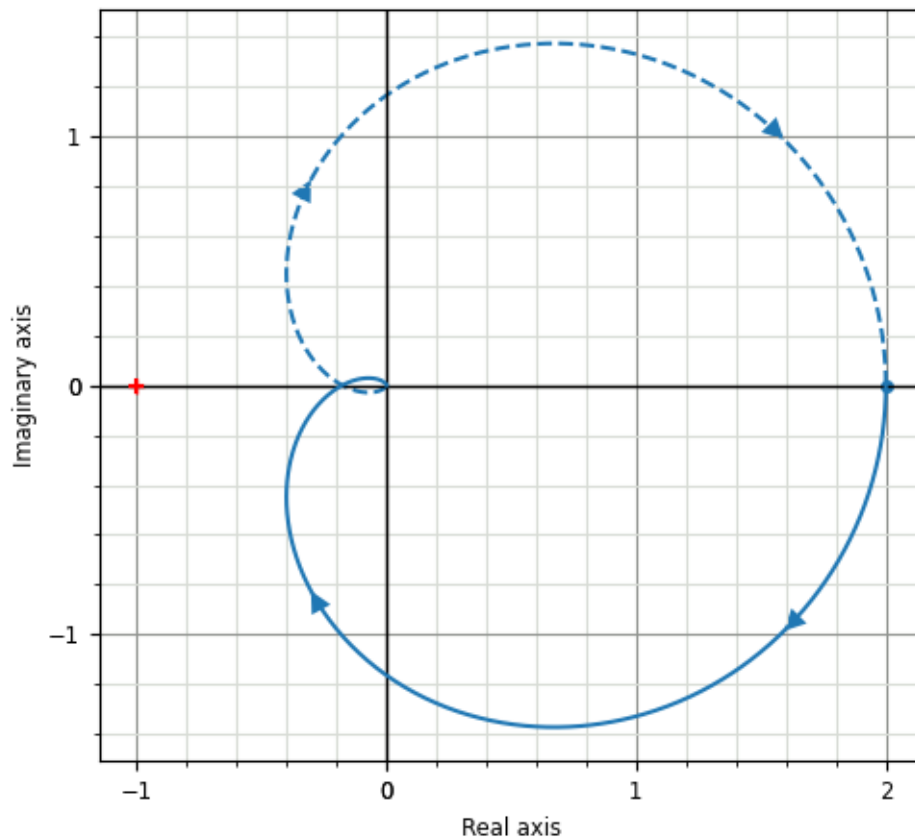
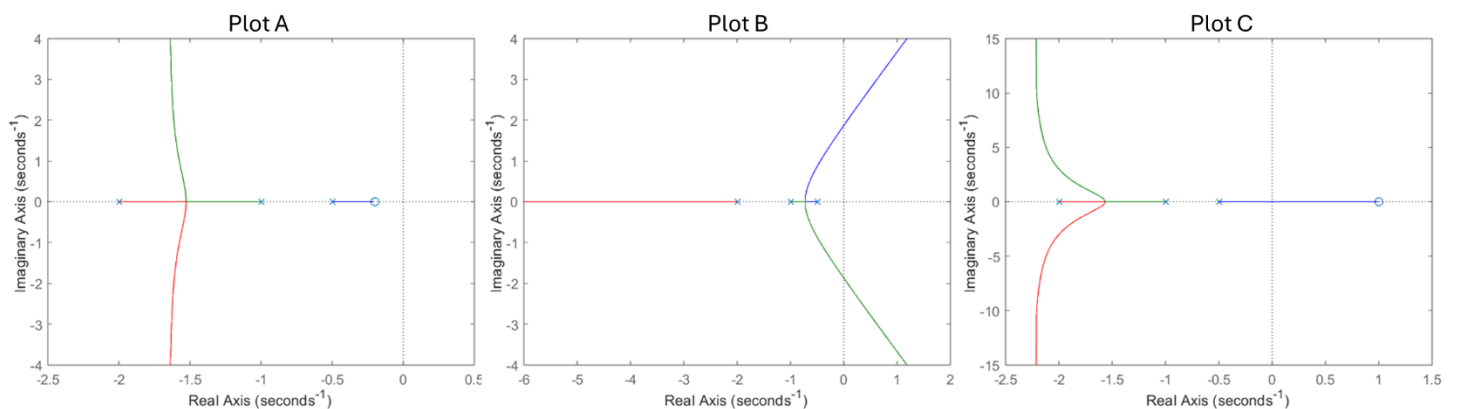


Tut Test 4 [10 marks]

The Nyquist plot of an unknown third-order system is shown below. The system is stable in open loop.



A) Three possible root locus plots for the system are shown below. Which one do you believe matches the system? Explain your reasoning with reference to the Nyquist plot. [3 marks]



B) Using the Nyquist plot, confirm that the DC gain of the uncontrolled open-loop system is 2. (You may draw on the plot to support your answer) [2 marks]

C) If you placed this system in negative feedback with a proportional controller, would it be possible to achieve tracking of a constant setpoint with 90% accuracy? Explain. [5 marks]

Solutions:

A) Plot B.

From the Nyquist plot, we see that it is possible for -1 to be encircled twice clockwise if the gain increases. The open loop system is stable so there are no RHP open loop poles, meaning that two clockwise encirclements would produce two unstable closed loop poles:

$$N = P_{cl} - P_{ol} \rightarrow 2 = P_{cl} - 0 \therefore P_{cl} = 2$$

We are therefore looking for a root locus where the closed-loop system goes from having zero to two unstable poles. The only one fitting this description is plot B.

B) The DC gain is a positive, real value, so it will be given by the point where the Nyquist plot intersects the positive real axis. This is at 2.

C) For the system to track the setpoint with 90% accuracy, we need the final value for a step input to be >0.9 :

$$\lim_{s \rightarrow 0} s \frac{1}{s} \frac{KG(s)}{1 + KG(s)} \geq 0.9 \therefore \frac{KG(0)}{1 + KG(0)} \geq 0.9 \therefore \frac{2K}{1 + 2K} \geq 0.9 \therefore K \geq 4.5$$

[2 marks]

To find out if this is possible, we need to check whether this gain is within the gain margin of the system. We can find the gain margin from the Nyquist plot by evaluating how much the plot can be expanded before the intersection with the negative real axis crosses -1. The intersection is at -0.2, so it can be multiplied by 5 before the closed-loop system becomes unstable. [2 marks]

The required gain of 4.5 is therefore within the gain margin, so tracking with this accuracy is possible. [1 mark]